

Chapter 4 — Measurement

How you know any of Chapter 3 worked

Chapter 3 ended owing you two answers. This is how you collect a debt like that from a system nobody can open.



Why prompting advice circulates as folklore

- Someone changes a prompt, the output looks better, and they tell you.
- They ran it once. They looked at the output before deciding what "better" meant. They did not run the old prompt again.
- Every one of those is a defect you would catch instantly in a colleague's experiment.

This room already has this skill. You do not know it applies here, because nobody has called a prompt an experimental condition in front of you before.



The anchor: an expert, a denominator, and a factor of two

An expert lecture defines a compute-time equation whose denominator is multiply-accumulates per second.

- Substitute a spec-sheet figure in FLOP per second and the result is wrong by exactly two.
- One MAC is two FLOP. The numerator needs the factor, or the denominator does.
- It was caught by an outside reader, in a comment thread, after publication.

A unit-definition error, in expert material, found by review, producing a clean factor of two. Everything this chapter teaches is in that sentence.



What the erratum actually teaches

- The expert was not careless. The equation is correct as written; the error appears when someone substitutes a number defined differently.
- Fluency did not help. The lecture is lucid, and the error survived it.
- Review worked — but only because the denominator's definition was written down where a reader could check it.

You cannot check a claim whose terms are undefined. That is why the pass criterion in this chapter comes before the output, not after.



The minimal eval

- Five fixed cases from your own work, written down and frozen.
- A pass criterion per case, defined before you look at any output. Binary. Checkable without seeing the other condition.
- Run condition A. Run condition B. **Count.**

Full specification, including the criterion-writing rules, is in `assets/eval-worksheet/README.md` . You take a blank copy home.



Choosing the five cases

- They must sit where the model is right some of the time. A case it always passes, or always fails, has zero power to detect anything, at any sample size.
- So calibrate first: write ten candidates, run each three times at baseline, discard anything scoring 3/3 or 0/3, keep the five nearest the middle.
- That costs about thirty runs and it is the difference between an eval that can detect something and one that cannot.

TODO(instructor) — the five cases, in mechanics, concrete or fluids, with a binary pass criterion each. Not written here: a case invented by someone who does not know the material comes out either trivially right or arguably wrong, and both destroy the exercise.



The conditions

A	baseline prompt
A'	the same baseline prompt, run again
B1	baseline + an expert persona
B2	baseline + an explicit instruction to work step by step
B3	baseline with extended thinking off

- A' is not a typo. It is the most important row in the table.
- B3 exists because the default configuration already reasons — without it, a null result on B2 has two readings and you cannot tell which.



Why A' exists

- Output is stochastic. Chapter 1 said so; you watched it in the temperature demo.
- One run per condition confounds the prompt effect with run-to-run spread.
- **A' measures the spread with nothing changed.** If A and A' disagree on two of five cases, an A-versus-B difference of two or fewer is inside the noise.

This is a repeatability check on an instrument. You would not accept a tank result without one.



Blinding, and what it does not achieve

- Strip the condition labels, rename outputs to opaque IDs, shuffle, and grade pass/fail only. Unblind afterwards.
- Discard the thinking transcript at capture — it is present under some arms and absent under B3, which identifies that arm with certainty.
- Insert a few duplicate outputs. They measure **grader** drift, which A' does not.

It is not a blind against the person who designed the conditions. An answer produced under *work step by step* tends to arrive in numbered steps — the manipulation and its marker are nearly the same thing. Closing that needs a second grader, and there isn't one. So it gets disclosed, not solved.



What five cases can support

Paired sign test, ties dropped:

Cases	Best possible split	Two-sided p
5	5–0	0.0625
6	6–0	0.0313
10	10–0	0.0020
10	9–1	0.0215
10	8–2	0.1094

- Five cases **cannot** reach two-sided $p < 0.05$ under any outcome. Six is the minimum.
- Power at $\alpha = 0.05$ is therefore **exactly zero** — combinatorial, not an estimate.



And how weak the best case really is

Take a true effect of +20 percentage points against a 50% baseline — the most favourable assumption available.

- Only about 2.5 of the five cases are expected to be informative. The rest tie.
- *No case contradicts B* occurs with probability 0.41 under that true effect and 0.21 under no effect. Likelihood ratio 2.0 — and that outcome is mostly ties, so it is not the best result available.
- The genuinely best result is the 5–0 row in the table opposite: five informative cases, all favouring B. Likelihood ratio 5.4.
- But a real +20-point effect produces that outcome only 0.5% of the time. The design's best case is both weak evidence *and* almost never obtained.
- For 80% power you would need 103 cases — exactly, by the same sign test. The normal approximation says 93; the discreteness of the exact test costs the other ten.

Say this **before** showing any result. This room is made of experimentalists; stating the limit first is what buys credibility for everything after.



So what is five cases for?

- A screen. It tells you whether an effect is large enough to justify a bigger eval.
- A noise floor. You learn how much your own instrument moves when nothing changes.
- A habit. You will never again say a prompt is better without knowing what you compared it to.

That is more than almost every prompting claim you will read this year has behind it.



Two more traps

- The noise floor is not a precise reading. Two disagreements out of five gives a 95% interval of roughly (0.05, 0.85). Treat the rule as a conservative screen, not an instrument.
- Three arms means three comparisons. And the per-arm rate cannot be 5%, because two slides ago you proved five cases have no 5% rejection region at all. At the only level actually attainable, 0.0625 per arm, three arms give a family-wise rate of about 18%. **Declare every arm**, including the boring ones.

Reporting only the interesting arm is post-hoc selection of the winning condition — and doing that in a chapter about method would be self-refuting.



The results

TODO(capture) — the instructor's eval, run in advance: five cases, five conditions, twenty-five graded outputs, with the noise floor read first and every arm declared.

Header block to be filled before the run, not after: model ID, surface, thinking setting and effort, date, fresh-session policy.

A demo where the folklore loses is worth more than one where it wins. So is a demo where five cases cannot tell — which is the most likely outcome and the most honest one.



Settling Chapter 3's first debt: the persona

- Chapter 3 was careful: the documentation claims a role focuses **behaviour and tone**. It never claimed accuracy.
- The published test of the accuracy claim: 162 roles, four model families, 2,410 factual questions. **Personas in system prompts did not improve performance** over no persona at all.
- **Its scope, in the source's own terms**: those model families, that question set. Not Opus 5, and not civil engineering. It is evidence against the folk claim, not a measurement of your task.

And the finding that belongs in *this* chapter rather than that one: picking the best persona **per question** does help — but identifying it in advance is **no better than chance**. That is post-hoc selection of the winning condition, and it is exactly what your own five cases will do to you if you let them.



Settling the second: think step by step

- The original result: eight worked chain-of-thought exemplars, on a 540-billion-parameter model, with the ability stated to emerge in sufficiently large models.
- That is a result about supplying worked examples. It is not a result about typing four words at a model that already reasons before it answers.

TODO(verify) — a re-evaluation reporting little benefit from explicit chain-of-thought on modern reasoning models is recorded at [P] and has not been read in full. It does not reach this slide until it has been. Your own eval is the evidence you actually own.



Record the version, or you have recorded nothing

- Model ID and date, beside every result. A prompt has a shelf life and Chapter 5 explains why.
- The settings that change behaviour: thinking, effort, fresh session or not.
- **Re-test before you rely on it again.** A working prompt is a measurement, and measurements expire.

Thread opened — verification. Prompts here, sources in Chapter 14, arguments in Chapter 17, and the whole logging discipline in Chapter 18.



Where this leaves us

- A method that fits on one page and turns a claim into a count.
- An honest account of what five cases cannot do, stated before the result rather than after.
- Two pieces of received wisdom, tested rather than repeated.

You can now measure whether something worked. Chapter 5 is about what breaks anyway.

